

BS Artificial Intelligence (BSAI)

Big Data Analytics

REPORT

AIMAN

F2023376387

Section : A2

Big Data Analytics: Project 1

From 2,000 social-media records to a marketing decision

Companion notebook: Project1_Social_Media_Analytics.ipynb (runs in Google Colab)

1. Data source, provenance and course concepts

Source: GitHub repository mrudukp/social-media-sentiment-analysis, file social_media_sentiment_raw.csv (2,260 rows, 13 columns, five platforms, 12 countries, Jan 2023 to Dec 2024).

Provenance and licence note.

- The repository has no LICENSE file, so reuse terms are not stated.
- The data is not verified as coming from Pakistani clothing brands. The post texts are short templated sentences, which suggests synthetic or heavily simulated data. This project teaches the workflow; it makes no claims about Pakistan.
- For Pakistan-specific conclusions, an authorised export or clearly labelled synthetic data is needed.

How the course topics appear in this project.

Course topic	Where it appears
What is Big Data?	2,000-row file compared with a 1,000x larger workload (Section 6)
Why Big Data?	Runtime, memory, file size and repeat-workflow cost measured at four sizes
Data sources / types	Posts, hashtags and engagement counts; numerical, categorical, text and date fields
EDA and data objects	Each row is one post; columns, missing values and duplicates inspected (Section 2)
Plots and statistics	Counts, mean/median, bar chart, histogram, box plot, scatter plot (Sections 3-4)
Analytics techniques	Classification, clustering and regression previewed (Section 7)

2. Data understanding and quality checks

The observational unit is one social-media post (not a user, comment or campaign).

column	type	meaning	missing values?
post_id	Identifier	One post (observational unit)	No
timestamp	Date/time	When the post was published	No
platform	Categorical	Twitter, Instagram, Facebook, TikTok, LinkedIn (case standardised)	No
user	Text/ID	Anonymised handle	No
followers_count	Numerical	Author's followers; stored as text with commas in raw file	Yes
verified	Categorical	Yes/No account verification	Yes
text	Text	Post content (templated sentences)	No

hashtags	Text	Hashtags; blank = no hashtags	Yes (means none)
country	Categorical	Author's country (whitespace/case standardised)	Yes
likes	Numerical	Likes; negative values invalid -> missing	Yes (invalid)
retweets_shares	Numerical	Shares / retweets	No
comments	Numerical	Comments; blank means not collected, not zero	Yes
sentiment	Categorical	Positive / Neutral / Negative label	Yes

Quality checks on the raw file

- Shape: 2,260 rows x 13 columns.
- Missing cells: hashtags 788, country 126, followers_count 90, comments 46, verified 44, sentiment 33. All other columns are complete.
- Exact duplicate rows: 44. Repeated post_id: 60.
- Negative likes (invalid): 8.
- The platform column holds 10 spellings for 5 platforms (for example Twitter and TWITTER); country has the same problem plus trailing spaces; followers_count is stored as text with commas.

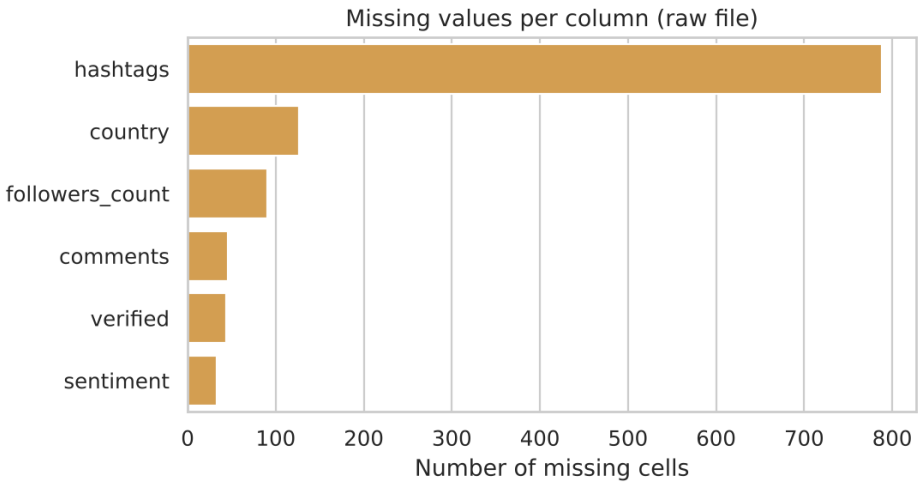


Figure 1. Missing values per column in the raw file.

Cleaning decisions

- Standardised case and whitespace, and converted followers_count to a number.
- Blank hashtags were treated as 'no hashtags' (a genuine zero). Blank comments were left missing: 'not collected' is not the same as zero. Negative likes were set to missing, not zero.
- Removed 60 duplicate posts after standardising. Only 44 were exact duplicates in the raw file; the other 16 were hidden by the case and comma differences. Analysis rows: 2,200.

3. Summary statistics

Platform	Posts (n)	Likes mean	Likes median	Comments median	Engagement mean	Engagement median
TikTok	390	47.35	44.0	4.0	60.32	56.0
Instagram	595	39.01	37.0	3.0	50.18	46.0
Twitter	726	34.04	32.0	3.0	43.39	40.0
Facebook	370	28.32	26.0	2.0	36.24	34.0
LinkedIn	119	21.22	20.0	2.0	26.84	24.5

Engagement = likes + shares + comments per post.

- Likes: mean 36.09, median 33, skewness 1.32, maximum 126. The mean sits above the median because of a right-skewed tail.
- The top 10% of posts account for 20.6% of all engagement, so a handful of strong posts pull averages up; medians are the safer summary.
- Likes and comments move together (Spearman rho = 0.59, n = 2,148).

4. Charts and interpretation

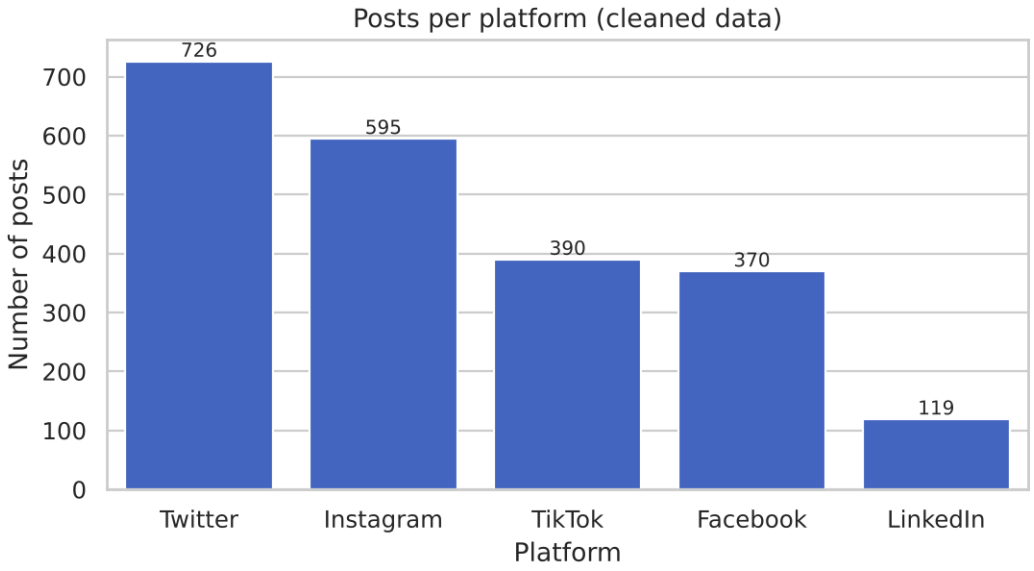


Figure 2. Posts per platform. Twitter and Instagram dominate; LinkedIn has only 119 posts.

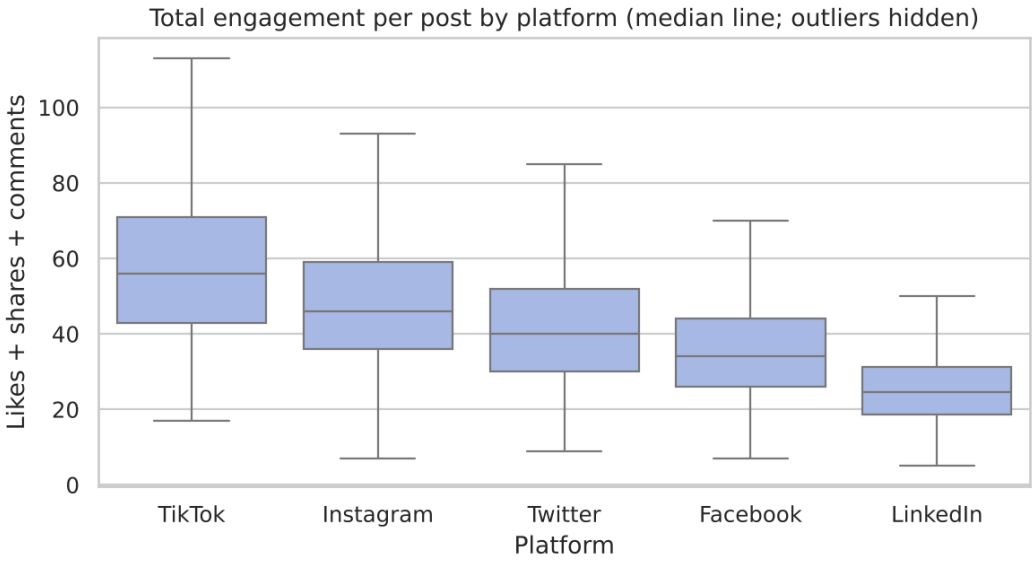


Figure 3. Engagement per post by platform. Medians fall from TikTok to LinkedIn.

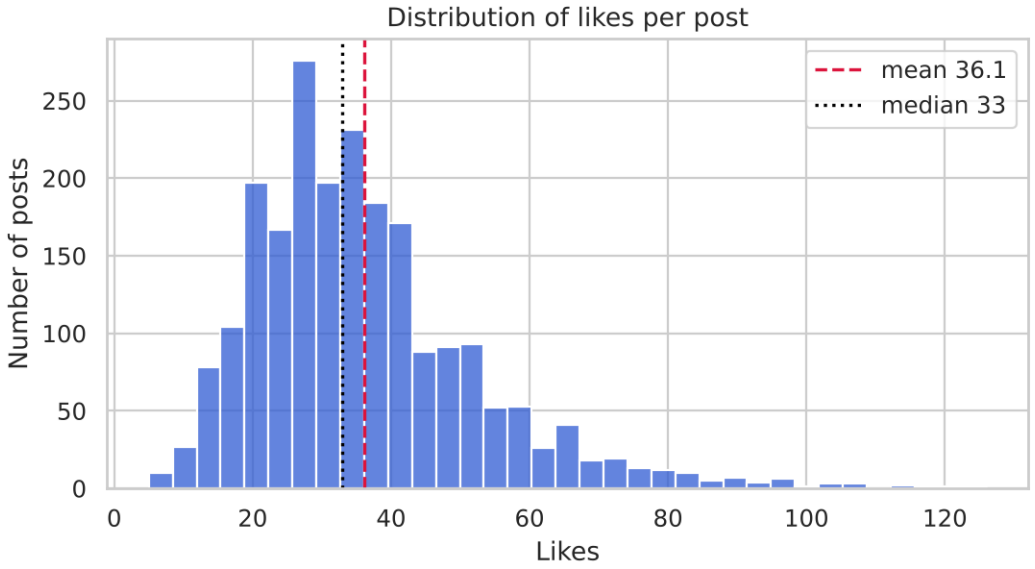


Figure 4. Likes are right-skewed: the mean (dashed) is above the median (dotted).

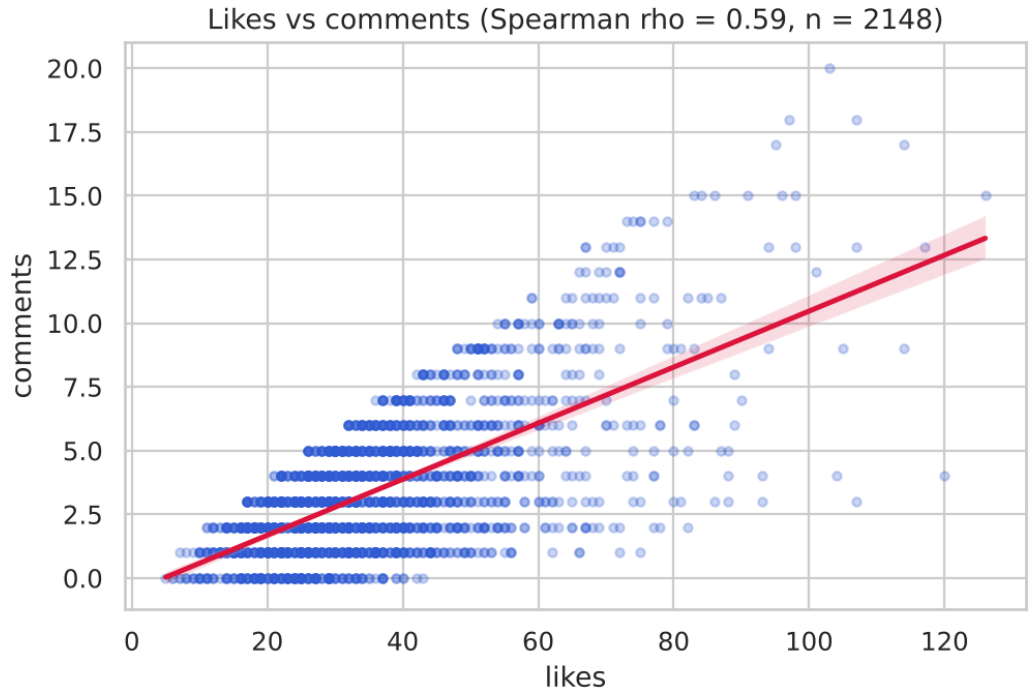


Figure 5. Likes and comments are positively related.

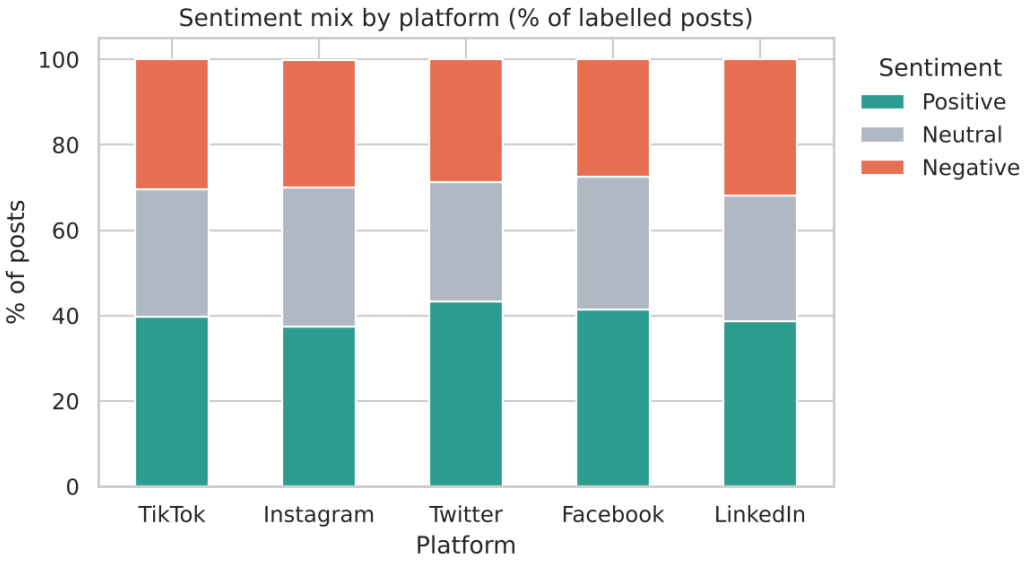


Figure 6. Sentiment mix is similar across platforms (about 38-43% positive).

5. Evidence-based observations and recommendation

Item	Detail
Finding 1	TikTok posts have the highest typical engagement and LinkedIn the lowest.

Evidence 1	Median engagement: TikTok 56, Instagram 46, Twitter 40, Facebook 34, LinkedIn 24.5. Kruskal-Wallis across platforms: H = 419.52, p < 0.001. Group sizes: 390, 595, 726, 370, 119.
Finding 2	Positive-tone posts earn more engagement than neutral or negative posts; negative posts beat neutral ones.
Evidence 2	Median engagement: Positive 50 (n = 879), Negative 40 (n = 636), Neutral 35 (n = 652); Kruskal-Wallis p < 0.001.
Action	Recommendation: a brand could test moving a share of content effort toward TikTok and toward clearly positive-tone posts for one campaign cycle, and compare results using one consistent metric, rather than assuming the pattern holds.
Limitation	No reach or impressions column, so this is engagement per post, not engagement rate or ROI. The data is unverified, probably synthetic, and the differences are descriptive, not causal. LinkedIn has a small sample (n = 119).

6. Scaling experiment and tool justification

The cleaned 2,200-row file was repeated to reach 1x, 10x, 100x and 1,000x workloads. This is a scaling experiment, not new real-world evidence. Each size ran in a fresh process on a 2-core, 8 GB machine. Operations: load the CSV, group-by mean, value counts, text-length mean, sort, and drop duplicates.

Rows	Workload	File (MB)	Load (s)	Analysis (s)	Dedupe (s)	Peak memory (MB)
2,000	1x	0.4	0.007	0.008	0.004	71.6
20,000	10x	3.7	0.049	0.016	0.019	88.0
200,000	100x	36.8	0.404	0.119	0.21	153.0
2,000,000	1000x	368.0	3.755	1.249	2.097	852.6

What this shows

- Cost grows roughly in proportion to rows: loading went from 0.007 s to 3.755 s and peak memory from 72 MB to 853 MB.
- Pandas did not fail at 2 million rows on this machine. There is no universal row count at which it fails; the need for Spark depends on data size, the operation, the hardware and service requirements.
- Where it could start to hurt (extrapolation, not measured): the 368 MB file already needs about 850 MB of memory, so if memory use keeps scaling linearly, a file 10x larger (about 20 million rows) would need roughly 8-9 GB, which is more than many laptops can spare, and repeated daily reprocessing would become slow.

Tool justification: a scalable tool such as Spark is worth investigating when volume, processing time, memory limits or distributed-storage needs justify it, for example full social-media archives across many brands. It is not needed for this 2,200-row file.

7. Preview of analytics techniques

A short conceptual demo only. Safeguards: train/test split (75/25), no engagement columns used as predictors of engagement (avoids leakage), and scaling before clustering.

Technique	Setup	Result (test set)
Classification	High vs low engagement (total engagement above the median of 42)	Accuracy 0.79 / F1 0.78 vs majority baseline 0.508
Regression	Predict likes from platform, followers, verified, sentiment, hashtags, hour	R2 0.683; MAE 7.14 likes vs 12.05 for always guessing the median
Clustering	K-means (k = 3) on scaled likes, shares, comments	Groups of 1173, 781 and 194 posts: low, mid and high engagement

Exit question: which technique helps the marketing question? Classification helps decide whether a planned post is likely to be a strong performer; regression estimates how many likes to expect; clustering segments posts into engagement tiers to study what top-tier posts share. Evidence needed: real reach data, content features beyond templated text, and a held-out time period.

8. Limitations

- Unverified, likely synthetic data with no licence; not representative of Pakistani clothing brands.
- Organic engagement only. ROAS and paid-ad performance cannot be calculated from it.
- Models use 2,200 templated posts and show the workflow, not production-grade accuracy.